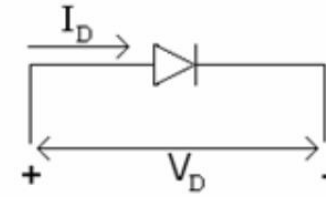


Diode equation



$$I_D = I_s (e^{V_D/nV_T} - 1)$$

where I_s is the reverse saturation current

V_D is the applied forward-bias voltage across the diode

n is an **ideality factor** which is a function of operating conditions and physical construction; it has a range between 1 & 2.

($n=1$ will be assumed unless otherwise noted)

$$V_T = \frac{kT}{q}$$

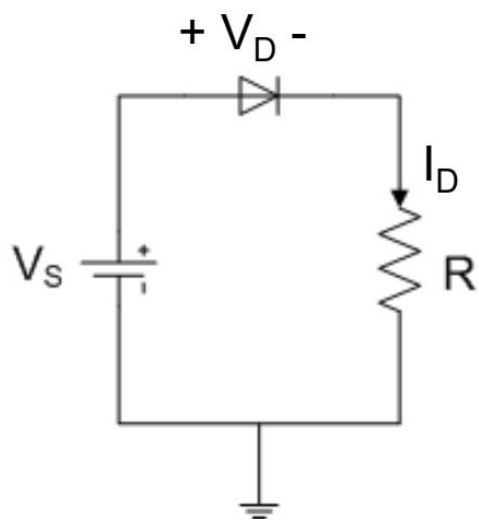
k is Boltzmann constant $= 1.38 \times 10^{-23}$ J/K

T is the absolute temperature in Kelvins

q is the electronic charge $= 1.6 \times 10^{-19}$ C

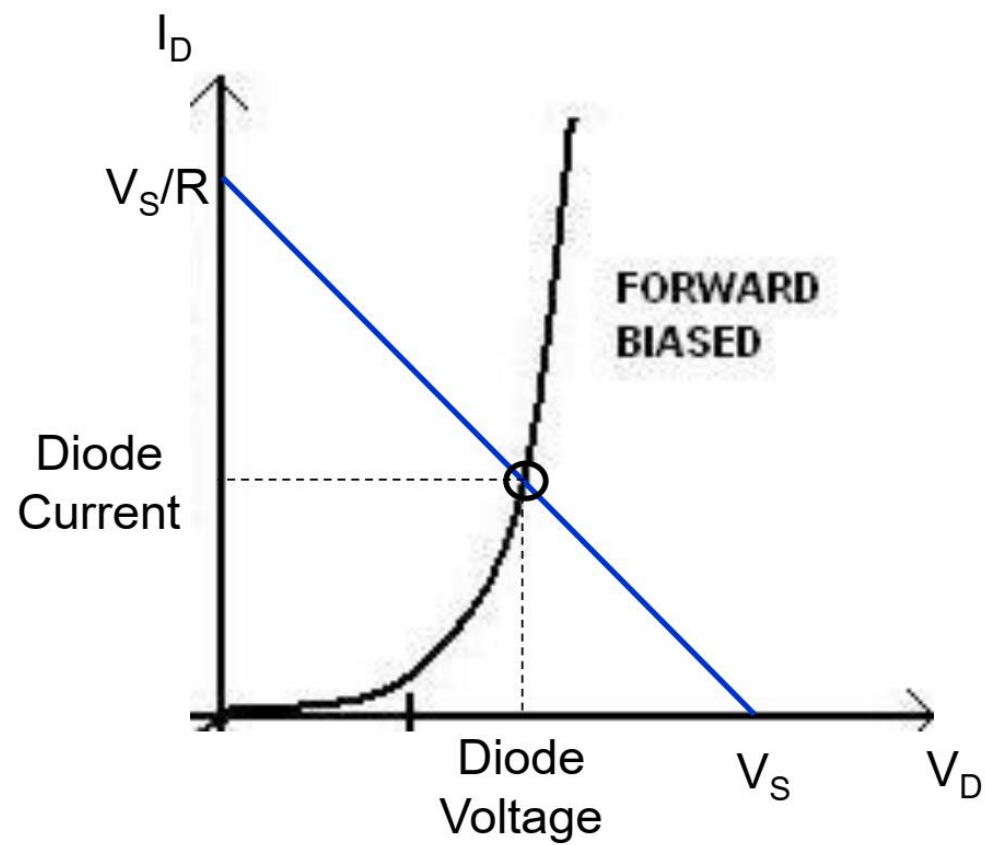
$V_T \approx 26$ mV at room temperature

For $V_D > 0$, $I_D \approx I_s e^{V_D/nV_T}$ and for $V_D < 0$, $I_D = -I_s$



$$I_D = \frac{V_S - V_D}{R}$$

$$I_D = \left(-\frac{1}{R}\right)V_D + \left(\frac{V_S}{R}\right)$$

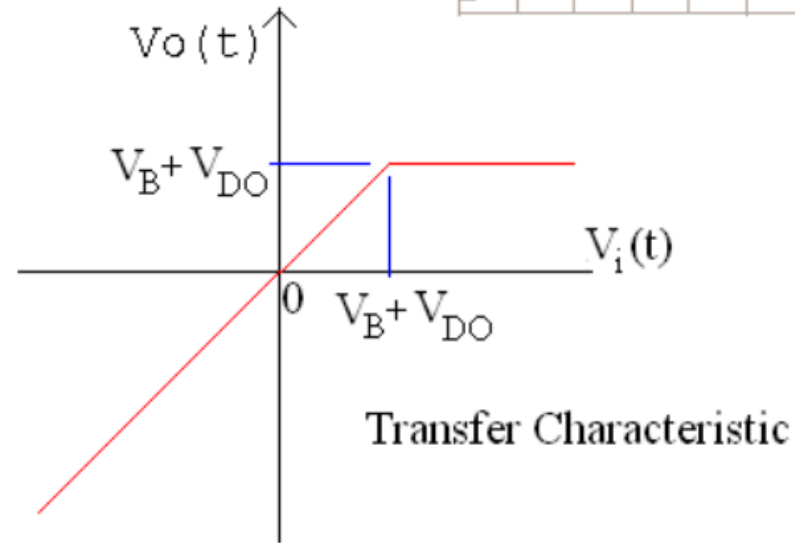
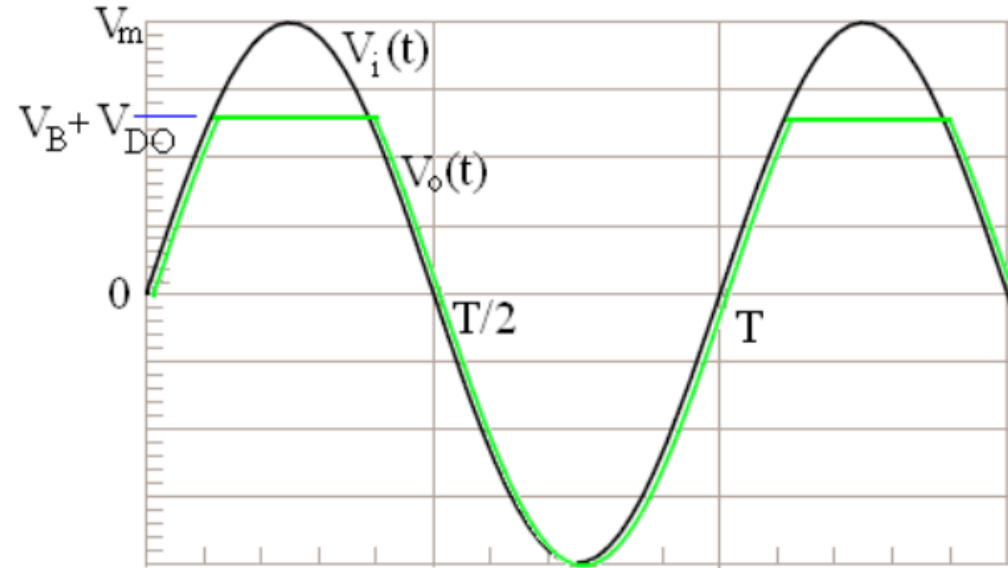
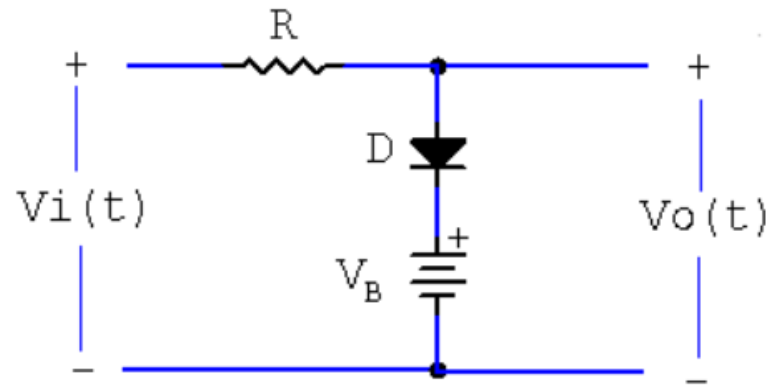


Wave Shaping Circuits

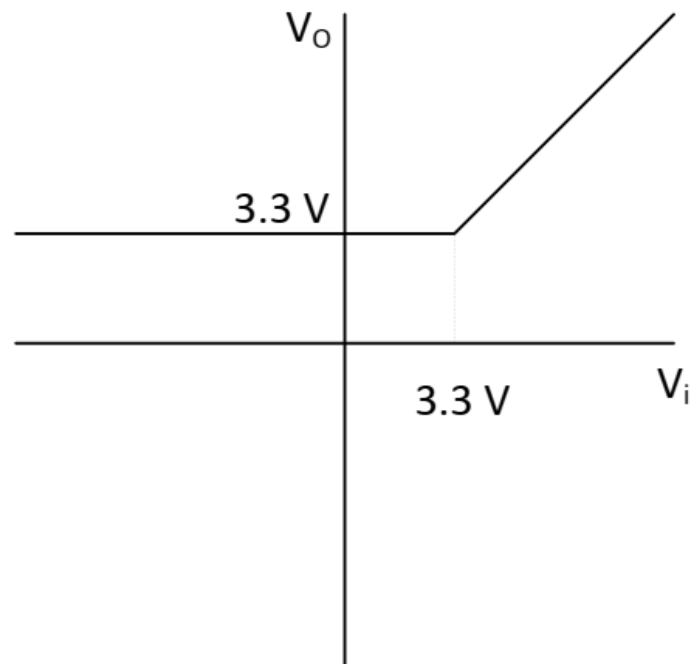
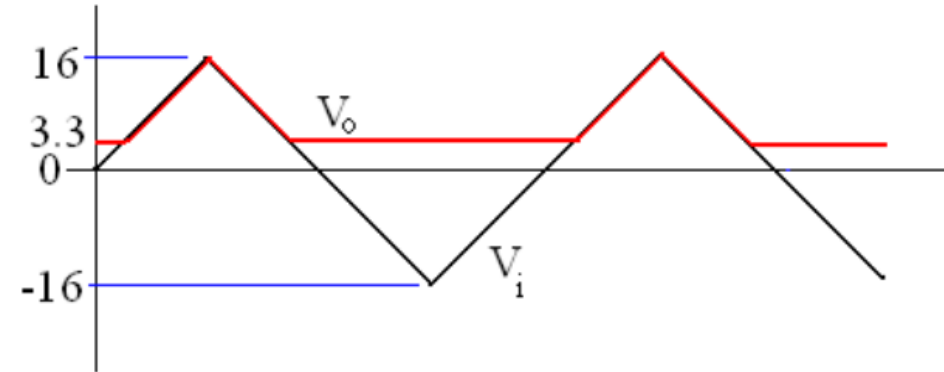
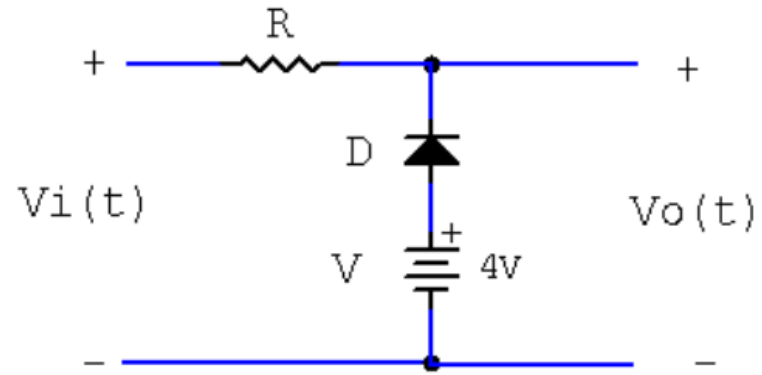
Diodes can be used in waveshaping circuits that either limit or “clip” portions of a signal, or shift the dc voltage level. These circuits are called **clippers** and **clampers** respectively.

Clippers are networks that employ diodes to clip away a portion of the input signal without distorting the remaining part of the applied waveform

Example 1: Clipper Circuit

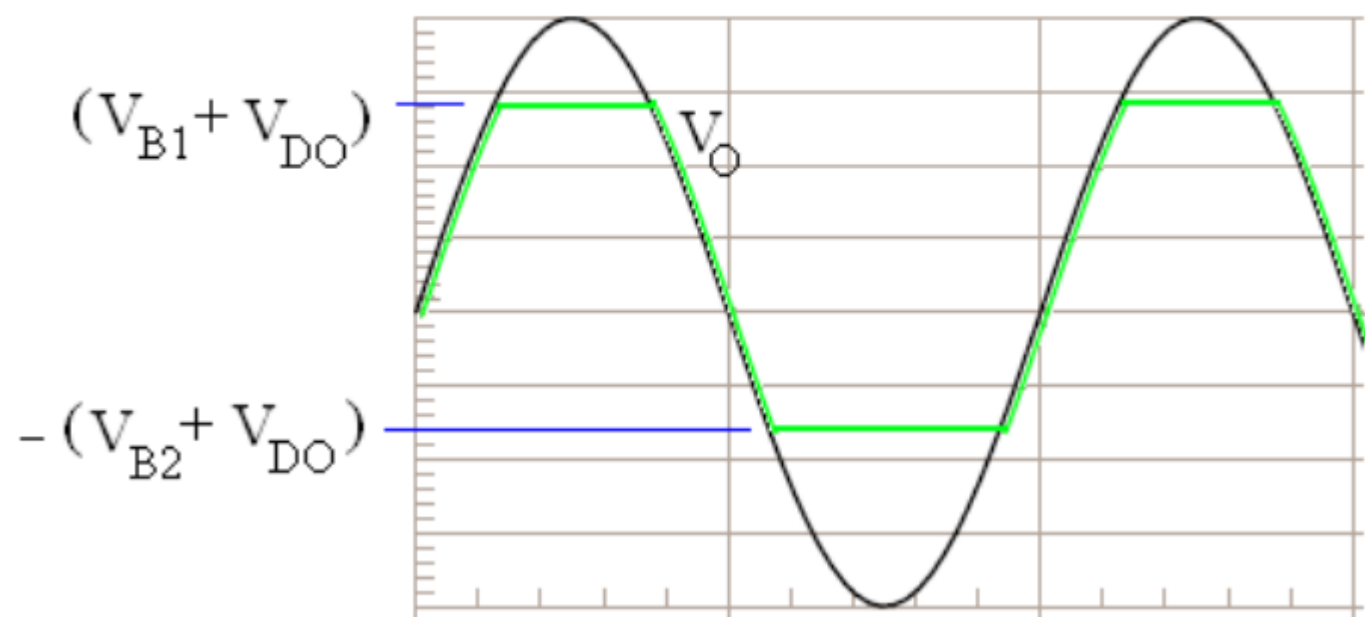
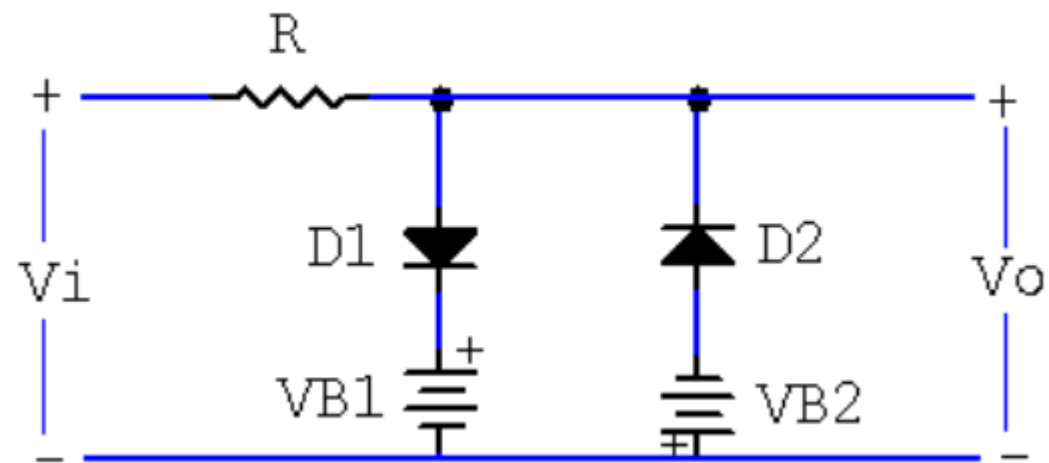


Example 2

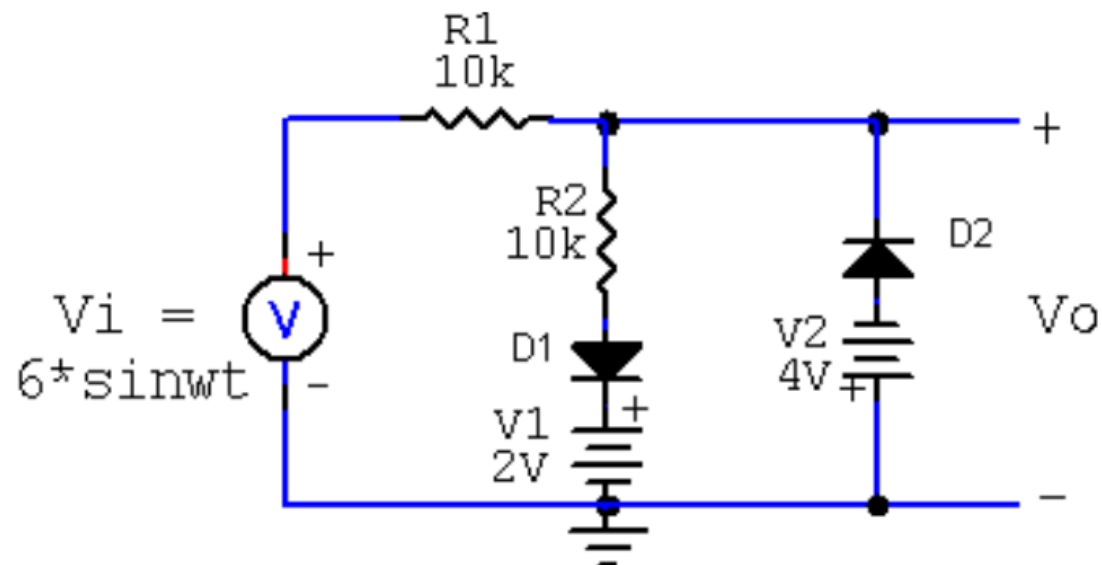


$$\begin{aligned} V_{DO} &= 0.7V \\ V_o &= (4 - 0.7)V \\ &= 3.3V \end{aligned}$$

Example 3

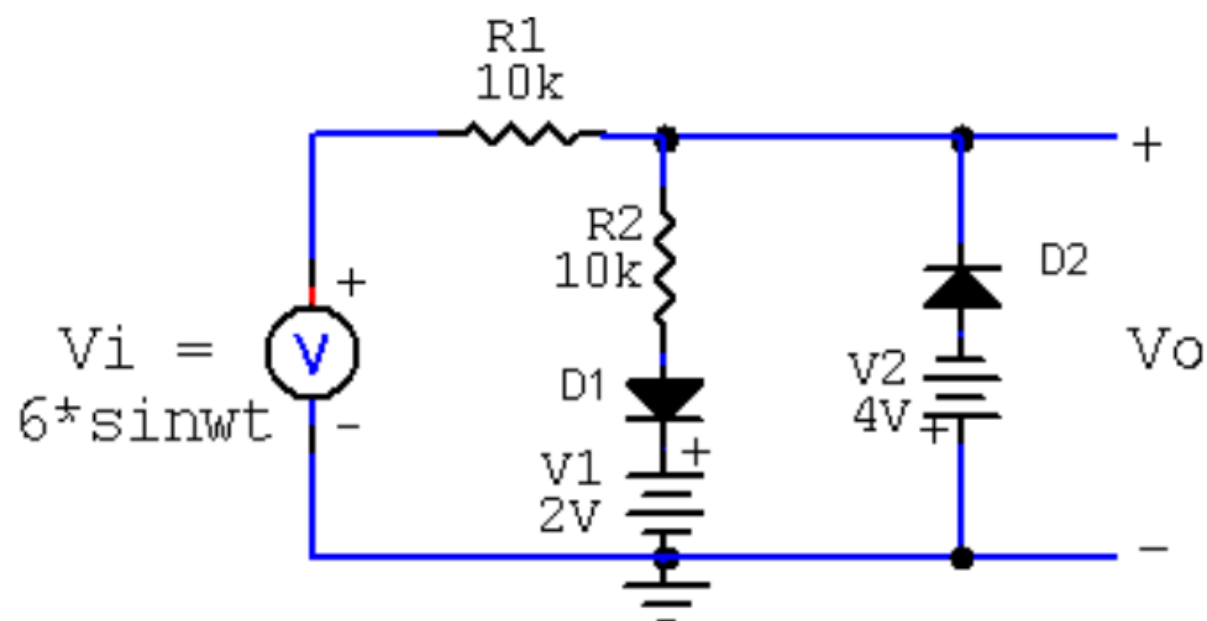


Example 4



Find V_o

Assume ideal diodes, i.e.
 $V_{D0}=0$, $R_f=0$ for both diodes.

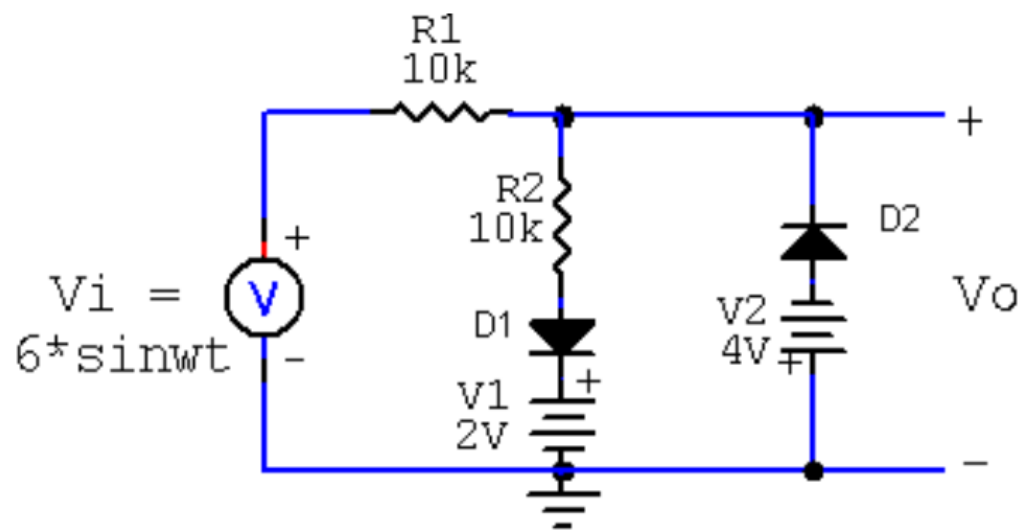


For $V_i < -4V$, D_2 turns ON and $V_O = -4V$

For $V_i > 2V$, D_1 turns ON and

$$i_1 = (V_i - 2)/(10+10)$$

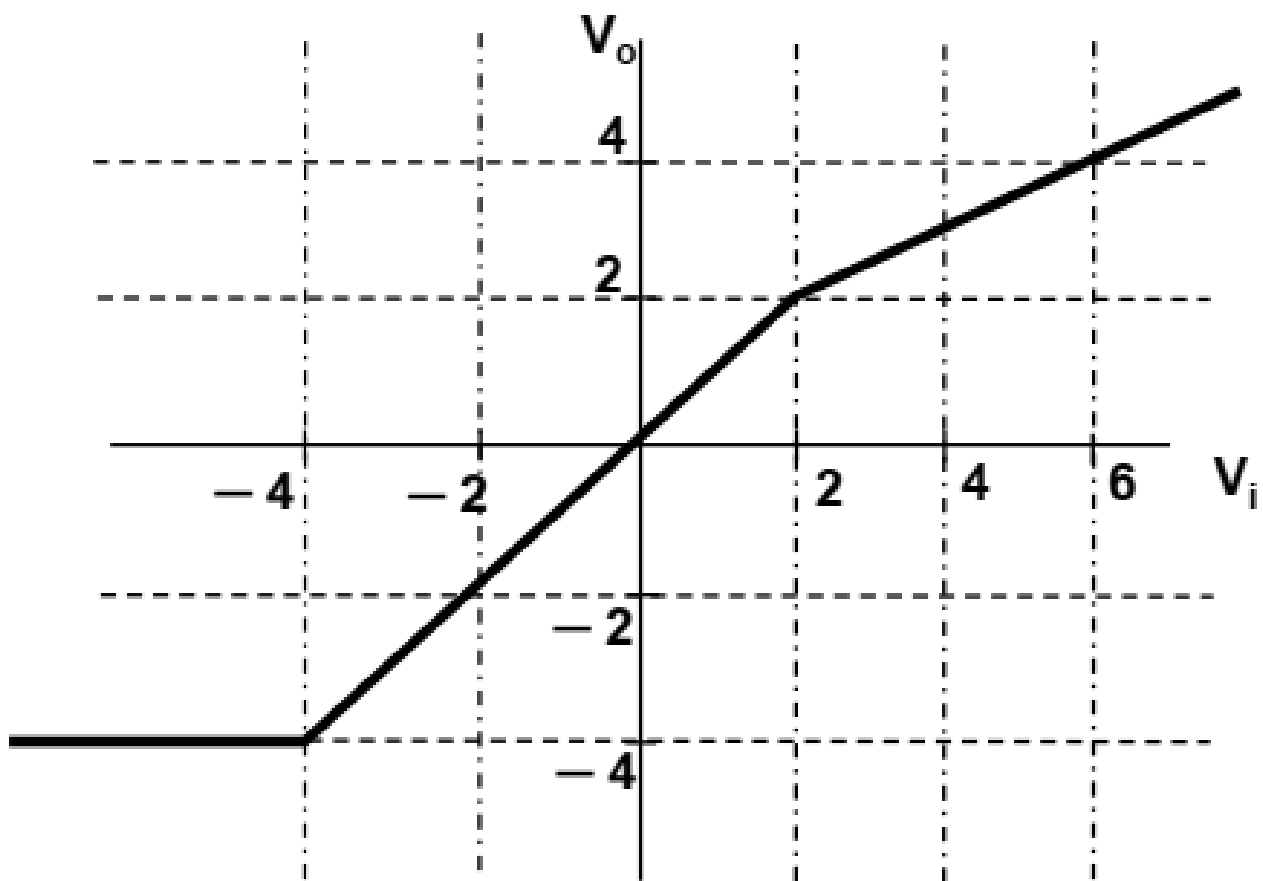
$$\begin{aligned} V_O &= i_1 R_2 + 2 = (V_i - 2)/20 \times 10 + 2 \\ &= V_i/2 - 1 + 2 = V_i/2 + 1 \end{aligned}$$

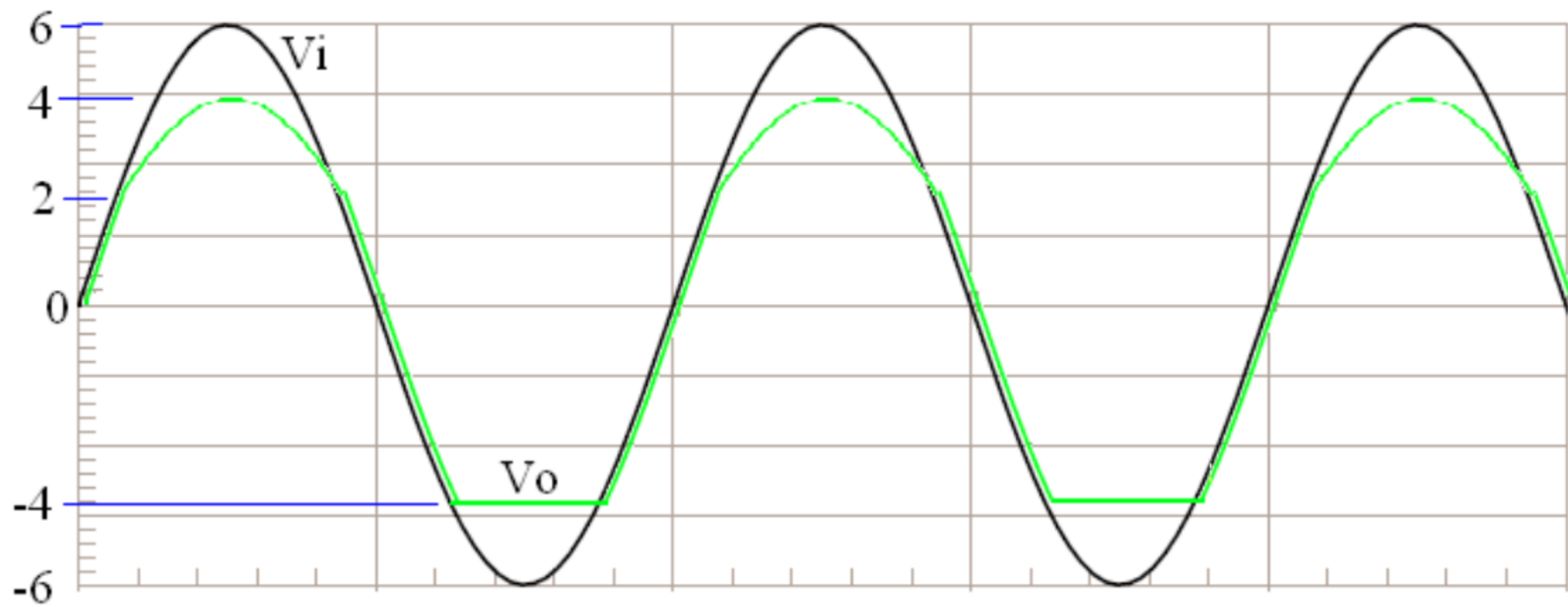


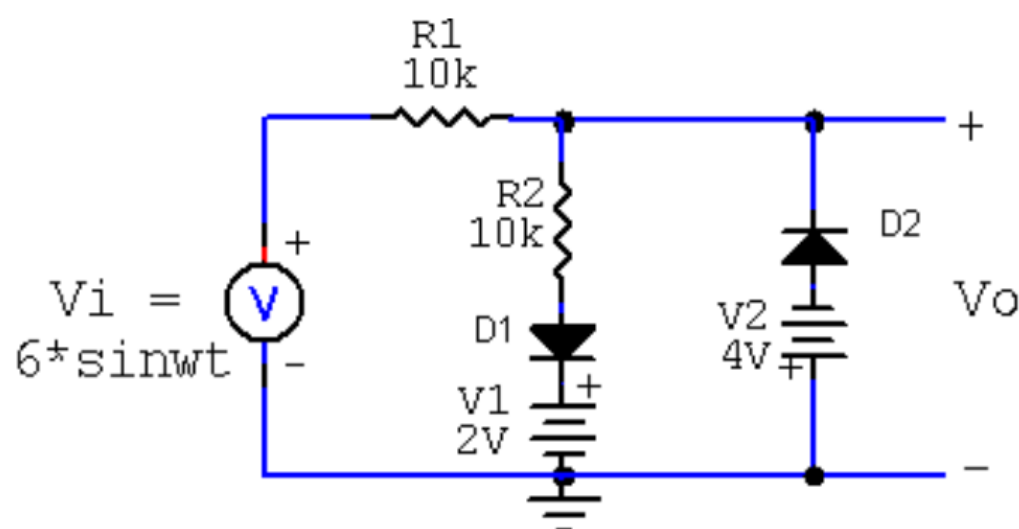
For $0 < V_i < 2V$, D_1 & D_2 are OFF as these are reverse biased.

Therefore, $V_o = V_i$

For $-4 < V_i < 0$, both D_1 & D_2 are OFF
& $V_o = V_i$





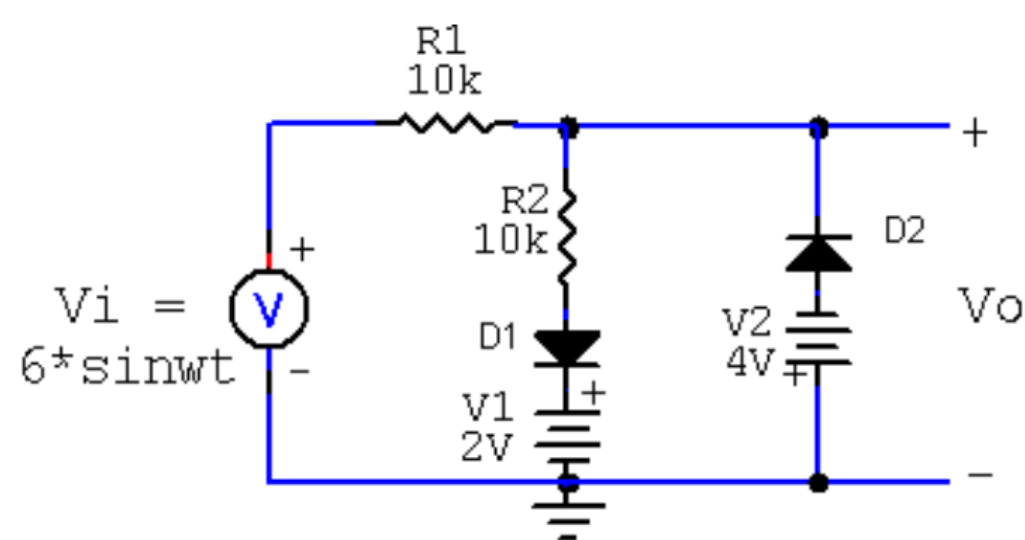


For $V_i < -4.7V$, D_2 turns ON and $V_o = -4.7V$

For $V_i > 2.7V$, D_1 turns ON and

$$i_1 = (V_i - 2.7)/(10+10)$$

$$\begin{aligned} V_o &= i_1 R_2 + 2.7 = (V_i - 2.7)/20 \times 10 + 2.7 \\ &= V_i/2 - 1.35 + 2.7 = V_i/2 + 1.35 \end{aligned}$$



For $0 < V_i < 2.7V$, D_1 & D_2 are OFF as these are reverse biased.

Therefore, $V_o = V_i$

For $-4.7 < V_i < 0$, both D_1 & D_2 are OFF
& $V_o = V_i$